

Software Requirements Specification

Sakar Robotics Fleet, Customer, and Inventory Management Platform

Prepared for business stakeholders, product management, design, and engineering teams

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1. Executive Summary

Sakar Robotics distributes, deploys, and supports Keenon commercial service robots through a multi-tier business network of distributors, sub-distributors, and direct clients. Today this business is coordinated through spreadsheets, email, and the Keenon vendor portal, which makes it difficult to track which organization owns which robot, who is authorized to see which customer's data, how spare parts and consumables are consumed across the network, and whether a robot is actually online and completing tasks.

This document specifies the requirements for the Sakar Robotics Fleet, Customer, and Inventory Management Platform (“the Platform”): a web-based system that models Sakar Robotics' business hierarchy, enforces strict role-based and organization-based access control, manages the robot fleet lifecycle end-to-end, tracks service inventory (consumables and spare parts), and integrates with the Keenon Open Platform to synchronize robot telemetry and dispatch tasks.

The Platform is designed around one non-negotiable principle: every organization in the network can see and act on its own data and the data of its authorized descendants — and nothing else. Distributors and sub-distributors must never be able to view, search, export, or modify data belonging to Sakar's direct clients, sibling distributors, or unrelated branches of the hierarchy. This is enforced structurally through organization-scoped data access, not merely through UI hiding.

1.1 Document Purpose

This SRS defines the functional and non-functional requirements, data model, access-control rules, Keenon integration contract, and delivery roadmap for the Platform. It is intended to align business stakeholders, product managers, UX designers, and engineers on a single source of truth before design and development begin.

1.2 Intended Audience

- Business stakeholders — validate that the platform reflects real Sakar Robotics business rules and revenue model.
- Product managers — use this document to scope releases, write backlogs, and prioritize the MVP.
- UX/UI designers — use the roles, workflows, and permission matrix to design role-appropriate screens.
- Developers and architects — use the data model, API integration, and security requirements to design the system.

2. Business Objectives

The Platform must deliver measurable business value in addition to functional correctness. Key objectives:

#	Objective	Description	Success Metric
O1	Centralize the sales/service hierarchy	Model Sakar Robotics, distributors, sub-distributors, and clients in one system of record instead of spreadsheets.	100% of active partners and clients onboarded within Phase 1
O2	Guarantee data isolation between partners	Prevent any distributor/sub-distributor from viewing another partner's customers, robots, or inventory.	Zero cross-tenant data-access incidents in audit review

#	Objective	Description	Success Metric
O3	Provide full fleet visibility	Give Sakar Admin real-time visibility into every robot's status, location, battery, and task history.	≥ 95% of active robots report live status via Keenon sync
O4	Reduce service downtime	Track spare parts and consumables so technicians can see stock availability before a service visit.	Reduce parts-related repeat visits by a target %, to be set with Ops
O5	Enable partner self-service	Let distributors and clients manage their own users, sites, and operators without Sakar involvement.	≥ 80% of partner-level admin actions performed without a Sakar support ticket
O6	Support business growth	Allow Sakar to onboard new distributors, sub-distributors, and clients without re-architecture.	Onboarding a new partner requires configuration only, no code changes
O7	Provide auditability and compliance	Maintain a complete, tamper-resistant audit trail of who did what, where, and when.	100% of create/update/delete/export actions logged and retrievable

3. User Roles and Organization Hierarchy

The Platform's access model is built on two intersecting concepts: the Organization Hierarchy (who reports to whom in the business network) and Roles (what a given user is allowed to do). A user's effective permissions are always the intersection of their Role and the Organization scope they belong to.

3.1 Organization Hierarchy

Sakar Robotics (Root Organization) └─ Sakar Admin / Sakar Staff (operate on behalf of the root organization)

└─ Client └─ Client (direct client of the Distributor, no sub-distributor layer) └─ Direct Client (client of Sakar Robotics with no distributor intermediary)

 └─ Distributor └─ Sub-distributor

A Client node under a Sub-distributor, a Client node directly under a Distributor, and a Direct Client node under Sakar Robotics are structurally identical records — only their parent organization differs.

3.2 Organization Node Reference

Node	Created / Managed By	Scope of Authority
Sakar Robotics (Root)	System-provisioned at deployment	Represents the company; not directly editable
Sakar Admin	Sakar Robotics (initial setup)	Full unrestricted access to all organizations, users, robots, inventory, and reports
Sakar Staff (Authorized)	Sakar Admin	Elevated staff granted permission to create Distributors and/or Direct Clients
Sakar Staff (Standard)	Sakar Admin	Operational/support staff with view access within an assigned scope; cannot create organizations
Distributor	Sakar Admin or Authorized Sakar Staff	Manages its own Sub-distributors and Clients only; no visibility into other Distributors
Sub-distributor	Its parent Distributor	Manages its own Clients only; no visibility into sibling Sub-distributors or the Distributor's other branches

Node	Created / Managed By	Scope of Authority
Client	Its parent Distributor or Sub-distributor	Manages its own internal users, sites, and robot operators only
Direct Client	Sakar Admin or Authorized Sakar Staff	Functionally identical to Client; reports directly to Sakar Robotics

3.3 Roles

Role	Organization Level	Primary Responsibilities
Sakar Admin	Sakar Robotics (root)	Full platform administration: users, all organizations, robots, inventory, Keenon integration config, global reporting
Sakar Staff — Authorized	Sakar Robotics (root)	Create/manage Distributors and Direct Clients per granted permission flags; assist with fleet and inventory setup
Sakar Staff — Standard	Sakar Robotics (root)	Read-only operational support within an assigned scope; escalates changes to Sakar Admin
Distributor Admin	Distributor	Manage own Sub-distributors, Clients, sites, users, inventory allocation, and reporting
Distributor Staff	Distributor	View-only operational role within the Distributor's own organization tree
Sub-distributor Admin	Sub-distributor	Manage own Clients, sites, users, inventory allocation, and reporting
Sub-distributor Staff	Sub-distributor	View-only operational role within the Sub-distributor's own organization tree
Client Admin (incl. Direct Client Admin)	Client / Direct Client	Manage own internal users, sites, robot operators, inventory consumption, and site-level reporting
Client User / Site Manager	Client / Direct Client	Day-to-day site operations: view robots, view/consume inventory, view site reports
Robot Operator	Client / Direct Client	Send permitted robot tasks at an assigned site; view assigned robot status
Technician	Any organization performing service	Perform service visits; consume spare parts/consumables; record service history

4. Functional Requirements

Requirements are grouped by module and prioritized using MoSCoW (Must / Should / Could).

4.1 Organization & Access Management

ID	Requirement	Priority
OAM-1	Support a hierarchical organization model with types Root, Distributor, Sub-distributor, Client, and Direct Client.	Must
OAM-2	Sakar Admin can create, edit, deactivate, and reactivate any organization or user account in the system.	Must

ID	Requirement	Priority
OAM-3	Sakar Admin can grant granular “Authorized Staff” permission flags to Sakar Staff accounts (e.g. “Can create Distributors”, “Can create Direct Clients”).	Must
OAM-4	Authorized Sakar Staff can create and manage only Distributor and Direct Client organizations, limited to their granted flags.	Must
OAM-5	A Distributor Admin can create and manage only Sub-distributor and Client organizations that are descendants of their own organization.	Must
OAM-6	A Sub-distributor Admin can create and manage only Client organizations that are descendants of their own organization.	Must
OAM-7	A Client Admin can create and manage only internal users, sites, and robot operators within their own organization.	Must
OAM-8	Every organization record stores an immutable parent-organization reference and a materialized hierarchy path used for scoped-query enforcement.	Must
OAM-9	The system prevents deletion of an organization that has active descendants, users, robots, or inventory, unless a cascading deactivation is explicitly confirmed by Sakar Admin.	Must
OAM-10	Organizations and users can be deactivated (soft-disabled) without losing historical records.	Should

4.2 User & Role Management

ID	Requirement	Priority
URM-1	Create a user with name, email, phone, organization, and role; send an email invitation with an expiring activation link.	Must
URM-2	Enforce a configurable password policy and support optional multi-factor authentication (MFA).	Must
URM-3	Role assignment is always scoped to the assigning user's own organization or an authorized descendant — a user can never be created in an organization the creator cannot access.	Must
URM-4	Users can view and edit their own profile (name, phone, password, MFA) but not their role or organization.	Should
URM-5	Sakar Admin and organization admins can suspend or reactivate a user account within their scope.	Must

4.3 Robot Fleet Management

ID	Requirement	Priority
RFM-1	Register a robot with model, serial number, Keenon robot ID, ownership organization, warranty period, and commissioning date.	Must
RFM-2	Assign a robot to a client organization and a specific customer site; reassignment is logged with before/after values.	Must
RFM-3	Track robot lifecycle status: In Stock, Allocated, Commissioned, Active, Under Service, Decommissioned, Lost/Damaged.	Must
RFM-4	Link a Sakar robot record to a Keenon robot ID; enforce a one-to-one mapping and reject duplicate links.	Must
RFM-5	Display live telemetry (status, battery level, location, current task) synchronized from the Keenon Open Platform.	Must
RFM-6	Maintain a full service history per robot, including parts consumed and technician notes.	Must

ID	Requirement	Priority
RFM-7	Sakar Admin can bulk-import robots via CSV/Excel upload.	Could

4.4 Customer & Site Management

ID	Requirement	Priority
CSM-1	Create and manage customer sites with name, address, geolocation, contact person, and operating hours.	Must
CSM-2	A site always belongs to exactly one client (or direct client) organization.	Must
CSM-3	A client organization may have multiple sites; each site has its own robot allocation and inventory context.	Must
CSM-4	View all robots and their status at a given site.	Must

4.5 Task Management

ID	Requirement	Priority
TM-1	Send a robot task (remote call, multi-point delivery, return-to-origin, or vertical-specific task) to a robot within the sender's authorized scope.	Must
TM-2	Display task status history: Queued, Dispatched, In Progress, Completed, Failed, Cancelled.	Must
TM-3	Restrict which task types an operator may send, based on role, site, and robot vertical (hotel, hospital, cleaning, elevator-enabled).	Must
TM-4	Record initiating user, organization, robot, site, and timestamp for every task for audit purposes.	Must

4.6 Inventory Management

ID	Requirement	Priority
INV-1	Maintain inventory records for robots, consumables, and spare parts, each associated with an owning organization.	Must
INV-2	Support stock transactions: Received, Allocated, Transferred, Installed, Consumed (service), Returned, Damaged/Lost, and Adjusted.	Must
INV-3	Support configurable reorder levels with automatic low-stock alerts.	Must
INV-4	Support periodic stock audits (cycle counts) with variance reporting.	Should
INV-5	Scope inventory visibility to the user's organization and authorized descendants; Sakar Admin has global visibility.	Must

See Section 8 for the full inventory data model, item categories, and transaction rules.

4.7 Keenon Open Platform Integration

ID	Requirement	Priority
KIN-1	Store and manage Keenon Open Platform credentials (client ID, client secret, access/refresh tokens, callback URL, IP whitelist) server-side only.	Must
KIN-2	Synchronize stores, robots, scenes, maps, points, robot status, battery, location, and task status from Keenon.	Must

ID	Requirement	Priority
KIN-3	Support remote call, multi-point delivery, return-to-origin, and vertical APIs (hotel, hospital, cleaning, elevator) where applicable to the robot model.	Must
KIN-4	Map every Keenon store ID and robot ID to exactly one Sakar client organization and customer site.	Must
KIN-5	Receive and process Keenon callbacks for live task and status updates, with signature validation and retry handling.	Must

See Section 7 for the complete Keenon integration specification.

4.8 Dashboards, Reporting, Security & Audit

ID	Requirement	Priority
RPT-1	Provide role-appropriate dashboards summarizing fleet status, tasks, and inventory within the viewer's scope.	Must
RPT-2	Provide exportable reports (CSV/Excel/PDF) scoped to the viewer's organization and authorized descendants.	Must
SEC-1	Enforce role-based access control (RBAC) combined with organization-based access control (OBAC) on every request.	Must
SEC-2	Record an immutable audit log entry for every create, update, delete, export, login, and permission change.	Must
SEC-3	Deny and log any attempt to access an organization outside the requester's authorized hierarchy scope.	Must

See Sections 9 and 10 for full reporting and security requirements.

5. Permissions and Authorization Matrix

The matrix below defines, per role, the level of access to each resource category. “Own + descendants” means the role’s own organization plus any organization structurally beneath it in the hierarchy that the role is authorized to manage — never a sibling or unrelated branch.

Symbol	Meaning
F	Full manage (create, edit, delete/deactivate), global (all organizations)
F*	Full manage, scoped strictly to own organization + authorized descendants
V	View only, scoped to own organization + authorized descendants
VG	View only, global (all organizations) — no edit rights
—	No access

5.1 Organization & User Management

Role	Sakar Staff Accounts	Distributor Orgs	Sub-distributor Orgs	Client Orgs	Sites	Own-Org Users
Sakar Admin	F	F	F	F	F	F
Sakar Staff — Authorized	—	F* (assigned)	V	F* (assigned, Direct Clients)	F* (assigned)	—
Sakar Staff — Standard	—	V	V	V	V	—
Distributor Admin	—	—	F*	F*	F*	F*
Distributor Staff	—	—	V	V	V	V
Sub-distributor Admin	—	—	—	F*	F*	F*
Sub-distributor Staff	—	—	—	V	V	V
Client Admin	—	—	—	—	F* (own)	F* (own)
Client User	—	—	—	—	V	V (limited)
Robot Operator	—	—	—	—	V	—
Technician	—	—	—	—	V	—

5.2 Fleet, Inventory, Reporting & System

Role	Robots & Keenon Mapping	Send Robot Tasks	Inventory	Reports & Dashboards	Keenon API Credentials	Audit Logs
Sakar Admin	F (global)	F (global)	F (global)	F (global)	F	F (global)
Sakar Staff — Authorized	F* (assigned)	V	F* (assigned)	VG	—	VG
Sakar Staff — Standard	V (assigned)	—	V (assigned)	VG	—	—
Distributor Admin	V*	—	F*	F*	—	V*
Distributor Staff	V*	—	V*	V*	—	—

Role	Robots & Keenon Mapping	Send Robot Tasks	Inventory	Reports & Dashboards	Keenon API Credentials	Audit Logs
Sub-distributor Admin	V*	—	F*	F*	—	V*
Sub-distributor Staff	V*	—	V*	V*	—	—
Client Admin	V* (own)	F* (delegated, own)	F* (own)	F* (own)	—	V* (own)
Client User	V* (own)	F* (if permitted)	V* / consume	V* (own)	—	—
Robot Operator	V (assigned)	F (permitted types only)	—	—	—	—
Technician	V (assigned)	—	F (consume/service, own org)	—	—	—

Note: "V*" for Robots & Keenon Mapping under Distributor/Sub-distributor reflects that Keenon store/robot mapping is created and owned by Sakar Admin (see Workflow 4, Section 12) — partners can view but not alter the mapping.

6. Data Model / Key Entities

Every user-facing entity is associated with an owning Organization, directly or via a parent entity, so that access control can be enforced consistently at the data layer.

Entity	Key Attributes	Relationships
Organization	id, name, type (Root/Distributor/SubDistributor/Client/DirectClient), parent_org_id, hierarchy_path, status	Self-referencing tree; parent of Users, Sites, Robots, Inventory
User	id, org_id, name, email, phone, role_id, status, mfa_enabled, last_login	Belongs to one Organization; has one Role
Role	id, name, scope_level, description	Referenced by User; associated with a set of Permissions
Permission	id, role_id, resource, action (create/read/update/delete/export)	Belongs to a Role; evaluated together with Organization scope
Site	id, client_org_id, name, address, geo_lat, geo_lng, contact_name, contact_phone, operating_hours, status	Belongs to one Client/Direct Client Organization; parent of Robot assignments
Robot	id, model, serial_number, keenon_robot_id, owner_org_id, assigned_client_org_id, assigned_site_id, warranty_start, warranty_end, commissioning_date, status, last_synced_at	Owned by an Organization; assigned to a Client Org + Site; linked 1:1 to a Keenon Robot Mapping
ServiceRecord	id, robot_id, technician_user_id, service_date, description, parts_consumed[], labor_notes	Belongs to a Robot; references StockTransactions for parts consumed
Task	id, robot_id, org_id, site_id, task_type, keenon_task_id, initiated_by_user_id, status, created_at, completed_at, payload	Belongs to a Robot and the initiating Organization/User
InventoryItem	id, category (Robot/Consumable/SparePart), sku, name, description, unit, reorder_level, owner_org_id	Owned by an Organization; subject of StockTransactions
StockTransaction	id, inventory_item_id, org_id, site_id, type (Received/Allocated/Transferred/Installed/Consumed/Returned/Damaged/Lost/Adjusted/Audited), quantity, from_org_id, to_org_id, reference_robot_id, performed_by_user_id, timestamp, notes	References an InventoryItem, an Organization,

Entity	Key Attributes	Relationships
		optionally a Robot and Site
KeenonStoreMapping	id, keenon_store_id, client_org_id, site_id, sync_status, last_synced_at	Maps one Keenon store to one Client Organization + Site
KeenonRobotMapping	id, keenon_robot_id, sakar_robot_id, store_mapping_id	Maps one Keenon robot to one Sakar Robot record
KeenonCredential	id, client_id, client_secret_encrypted, access_token_encrypted, refresh_token_encrypted, token_expires_at, callback_url, ip_whitelist, approved_permissions	Server-side only; never returned to any frontend client
AuditLog	id, actor_user_id, actor_org_id, action, resource_type, resource_id, before_state, after_state, ip_address, timestamp	References the acting User/Organization and the affected resource
Notification	id, user_id, org_id, type, message, read_status, created_at	Belongs to a User within an Organization

6.1 Key Relationships

- Organization is self-referencing (parent_org_id) and carries a materialized hierarchy_path (e.g. “/root/dist-12/subdist-04/client-31”) used to enforce scoped access in every query.
- Every Robot, Task, InventoryItem, StockTransaction, and Site record carries an organization reference — there is no orphaned or organization-less business data.
- A Robot is owned by Sakar Robotics (or a partner, if resold) but assigned to exactly one Client Organization and one Site at a time; historical assignments are preserved for audit.
- KeenonStoreMapping and KeenonRobotMapping form the bridge between Keenon's identifiers and Sakar's organization/robot records; they are maintained exclusively by Sakar Admin.

7. Keenon Open Platform API Integration

The Platform integrates with the Keenon Open Platform as a server-to-server backend integration. No Keenon credential, secret, or raw token is ever exposed to a browser or mobile client.

7.1 Credential & Connection Management

- Store client ID, client secret, access token, and refresh token encrypted at rest, accessible only to backend services.
- Maintain the approved store permissions granted by Keenon for this integration.
- Configure and validate the callback URL registered with Keenon.
- Maintain and enforce an IP whitelist for both outbound calls to Keenon and inbound callbacks from Keenon.
- Automatically refresh access tokens before expiry; alert Sakar Admin on refresh failure.

7.2 Data Synchronization

Data Type	Sync Direction	Notes
Stores	Keenon → Sakar	Synchronized store list is mapped to Sakar client organizations and sites
Robots	Keenon → Sakar	Matched to Sakar robot records via keenon_robot_id
Scenes / Maps / Points	Keenon → Sakar	Used to validate and configure multi-point delivery tasks
Robot status / battery / location	Keenon → Sakar (near real-time via callback + polling fallback)	Displayed on fleet and site dashboards
Task status	Keenon → Sakar (callback)	Drives the Task status history described in Section 4.5

7.3 Robot Task & Control APIs

- Remote call — summon a robot to a specified point.
- Multi-point delivery — dispatch a robot through an ordered sequence of points.
- Return-to-origin — recall a robot to its charging/home position.
- Vertical-specific APIs where applicable to the robot's deployment: hotel (room delivery), hospital (ward/sample transport), cleaning (zone cleaning routes), and elevator control/integration for multi-floor navigation.

Only task types enabled for a given robot model and site vertical are exposed to operators at that site (see TM-3).

7.4 Organization Mapping

- Every keenon_store_id is mapped to exactly one Sakar client organization and one customer site.
- Every keenon_robot_id is mapped to exactly one Sakar robot record, which in turn carries the owning and assigned organization.
- Unmapped stores or robots discovered during sync are queued in a “Pending Mapping” review list for Sakar Admin; they are not visible to any partner until mapped.

7.5 Callback & Webhook Handling

- Expose a single authenticated callback endpoint for Keenon to push live task and status updates.
- Validate the signature of every inbound callback before processing; reject and log unsigned or invalid requests.
- Persist every inbound callback payload and its processing outcome to an integration log for troubleshooting.
- Retry failed callback processing with exponential backoff; surface persistent failures to Sakar Admin.
- Fall back to scheduled polling of the Keenon API if callbacks are not received within an expected interval, to avoid stale fleet status.

7.6 Security Requirements for the Integration

- Keenon client secret, access token, and refresh token are never sent to, stored in, or logged by any frontend client or browser.
- All outbound requests to Keenon and inbound callbacks occur over TLS.
- Access to view or rotate Keenon credentials is restricted to Sakar Admin only (see Section 5.2).
- Integration logs are retained per the audit retention policy in Section 10.3 and are accessible only to Sakar Admin.

8. Inventory Management Requirements

8.1 Robot Inventory

Each robot record tracks the following attributes:

Attribute	Description
Robot Model	Keenon product model (e.g. delivery, cleaning, disinfection robot)
Serial Number	Manufacturer serial number, unique across the fleet
Keenon Robot ID	Unique identifier from the Keenon Open Platform, linked 1:1
Ownership	The organization that owns the physical asset (Sakar or reselling partner)
Assigned Client	The client organization currently operating the robot
Assigned Site	The specific customer site where the robot is deployed
Warranty	Warranty start and end dates
Status	In Stock, Allocated, Commissioned, Active, Under Service, Decommissioned, Lost/Damaged
Commissioning Date	Date the robot was activated at the client site
Service History	Chronological log of all service visits and parts consumed

8.2 Consumables

Tracked as a distinct inventory category with a lower unit value and higher turnover than spare parts.

Representative items:

- Brushes
- Filters
- Cleaning pads
- Cleaning solution
- Dust bags
- Batteries
- Other consumables as defined by Sakar Admin / catalog management

8.3 Spare Parts

Tracked as a distinct inventory category, typically higher-value, serialized where relevant. Representative items:

- Motors
- Wheels
- Sensors (LiDAR, ultrasonic, camera)
- Controllers
- Chargers
- PCBs
- Other repair items as defined by Sakar Admin / catalog management

8.4 Stock Transaction Types

Transaction Type	Description
Stock Received	New inventory received into an organization's stock, typically from Sakar or the upstream distributor
Allocation	Stock earmarked for a specific client, site, or robot without physical transfer
Transfer	Movement of stock between two organizations in the hierarchy (e.g. distributor to sub-distributor)
Installation	A spare part is installed into a specific robot during service
Service Consumption	A consumable or part is used up during a service visit
Return	Stock returned up the hierarchy (e.g. client to distributor) or to Sakar
Damaged / Lost	Stock written off due to damage or loss, with a required reason note
Stock Adjustment	Manual correction of recorded quantity, requiring a reason and approver
Reorder Level Trigger	Automatic alert generated when stock falls below the configured reorder level
Stock Audit	Periodic physical count reconciled against system quantity, producing a variance report

8.5 Visibility & Scoping Rules

- Inventory is visible only within the organizations the requesting user is authorized to access (own organization plus authorized descendants).
- Sakar Admin has full inventory visibility and management rights across all organizations.
- A Distributor cannot view inventory belonging to another Distributor, a Direct Client, or any organization outside its own descendant tree.
- Every stock transaction is recorded against the organization(s) involved (from_org_id / to_org_id) to preserve a complete chain of custody.

9. Dashboard and Reporting Requirements

9.1 Role-Based Dashboards

Role	Dashboard Focus
Sakar Admin	Global fleet health, org growth, cross-partner inventory levels, Keenon sync health, audit summary
Sakar Staff (Authorized/Standard)	Assigned-scope fleet and task summary, pending Keenon mapping queue
Distributor / Sub-distributor Admin	Own network's robot status, task volume, inventory levels, and reorder alerts
Client Admin / Client User	Site-level robot status, task history, and inventory consumption
Robot Operator	Assigned robots, task queue, and task history at their site
Technician	Assigned service visits, parts consumption, and service history

9.2 Standard Reports

Report	Description	Available To
Fleet Status Report	Robot counts by status, model, and organization	Sakar Admin (global), partners/clients (own scope)
Task Completion Report	Task volumes, success/failure rates, by robot/site/date range	Sakar Admin (global), partners/clients (own scope)
Inventory Stock Level Report	Current stock by item, category, and organization, with reorder flags	Sakar Admin (global), partners/clients (own scope)
Service History Report	Service visits, parts consumed, technician, and robot	Sakar Admin (global), partners/clients (own scope)
Organization Hierarchy Report	Snapshot of the full or scoped partner/client tree with counts	Sakar Admin (global), partners (own descendants only)
Keenon Sync Health Report	Sync success rate, callback failures, pending mappings	Sakar Admin only

9.3 Export & Scheduling

- Reports can be exported to CSV, Excel, and PDF, always scoped to the exporting user's authorized organizations.
- Every export action is recorded in the audit log, including the filter/scope applied.
- Users with report access may optionally schedule recurring report delivery via email within their own organization scope.

10. Security, Audit, and Data-Isolation Requirements

10.1 Role-Based Access Control (RBAC)

- Every action in the system is governed by a Role that defines the set of permitted operations (create/read/update/delete/export) per resource type.
- Roles are assignable only within the assigner's own authorized organization scope.

10.2 Organization-Based Access Control (OBAC) and Data Isolation

- Every data query is filtered by the requesting user's organization hierarchy path — a user may only retrieve records whose organization is their own or an authorized descendant.
- Distributors and sub-distributors must never view, search, export, modify, or otherwise access data belonging to Sakar's direct clients, other distributors, or unrelated branches of the hierarchy — enforced at the data-access layer, not only in the UI.
- Every API request is evaluated against both RBAC (can this role perform this action) and OBAC (does this organization scope include the target record) before being permitted.
- Attempts to access out-of-scope data return an authorization error and are recorded in the audit log with actor, target, and reason (see Workflow 7, Section 12).

10.3 Audit Logging

- Every create, update, delete, export, login/logout, permission change, and denied-access attempt is logged with actor, organization, action, affected resource, before/after state, timestamp, and source IP.
- Audit logs are append-only and cannot be edited or deleted by any role, including Sakar Admin, through the application layer.
- Audit logs are retained for a minimum period to be defined with compliance stakeholders (see Section 15) and are viewable per the scope defined in Section 5.2.

10.4 Authentication & Session Management

- Enforce strong password policy and support multi-factor authentication.
- Sessions expire after a configurable period of inactivity; tokens are invalidated on password change or account suspension.
- Support account lockout after repeated failed login attempts, with notification to the affected user.

10.5 Encryption & Secrets Management

- Data is encrypted in transit (TLS) and at rest for all sensitive fields, including Keenon credentials and tokens.
- Keenon client secret, access token, and refresh token are stored only in backend secret storage and are never exposed to any frontend client, log, or export.

11. Non-Functional Requirements

Category	Requirement	Target
Performance	Standard dashboard and list views load quickly under normal load	P95 page load $\leq 2.5s$; P95 API response $\leq 500ms$
Scalability	Support growth in partners, clients, robots, and inventory without redesign	Support $\geq 10,000$ robots and $\geq 5,000$ organizations at launch scale
Availability	Platform is available during business operations, including for time-sensitive robot tasks	$\geq 99.5\%$ monthly uptime for core services
Usability	Role-appropriate UI so partners and clients can self-serve without training	New client admin can complete first user/site setup unaided
Compatibility	Support current major desktop browsers and responsive mobile web for field/technician use	Latest 2 versions of Chrome, Edge, Safari, Firefox
Localization	Support future multi-language rollout as the partner network grows internationally	English at launch; architecture supports adding locales
Maintainability	New organization types, task types, or inventory categories can be added via configuration	No code change required to onboard a new partner or item category
Backup & Disaster Recovery	Protect against data loss	Daily backups; defined RPO/RTO to be confirmed with infrastructure stakeholders
Auditability	All security-relevant events are traceable	100% coverage per Section 10.3

12. User Workflows

12.1 Workflow 1 — Sakar Admin Creates a Distributor

Step	Actor	Action	System Response / Validation
1	Sakar Admin	Navigates to Organizations → Distributors → New Distributor	System displays the Distributor creation form
2	Sakar Admin	Enters distributor company details and an initial Distributor Admin user	System validates required fields and email uniqueness
3	System	Creates the Distributor organization as a child of Sakar Robotics with a hierarchy_path	New organization and admin user are persisted; invite email sent
4	System	Logs the creation event	Audit log entry created with actor = Sakar Admin, resource = Distributor

12.2 Workflow 2 — Distributor Creates a Sub-distributor

Step	Actor	Action	System Response / Validation
1	Distributor Admin	Navigates to Organizations → Sub-distributors → New Sub-distributor	System confirms the actor's role permits this action (OAM-5)
2	Distributor Admin	Enters sub-distributor details and an initial Sub-distributor Admin user	System validates fields; scopes the new org's parent to this Distributor
3	System	Creates the Sub-distributor as a descendant of the Distributor's hierarchy_path	New organization is only visible to this Distributor and Sakar Admin
4	System	Logs the creation event	Audit log entry created and scoped to the Distributor's organization

12.3 Workflow 3 — Distributor/Sub-distributor Creates a Client and Client Admin

Step	Actor	Action	System Response / Validation
1	Distributor or Sub-distributor Admin	Navigates to Clients → New Client	System confirms creator's role and scope (OAM-5/OAM-6)
2	Admin	Enters client company details, initial site, and Client Admin user	System validates fields and creates the Client as a descendant of the creator's organization
3	System	Sends activation invite to the Client Admin	Client Admin can log in and manage only their own organization thereafter (OAM-7)
4	System	Logs the creation event	Audit log entry scoped to the creating Distributor/Sub-distributor

12.4 Workflow 4 — Sakar Assigns a Robot to a Client Site and Links It to a Keenon Robot

Step	Actor	Action	System Response / Validation
1	Sakar Admin	Selects a robot record in In Stock or Allocated status	System shows eligible client organizations and sites

Step	Actor	Action	System Response / Validation
2	Sakar Admin	Assigns the robot to a Client organization and a specific Site; sets status to Commissioned	System updates assigned_client_org_id and assigned_site_id
3	Sakar Admin	Links the robot to its Keenon robot ID (from the synced Keenon robot list)	System creates a KeenonRobotMapping, enforcing 1:1 uniqueness
4	System	Begins telemetry sync for the robot	Status, battery, and location become visible to the assigned Client per Section 5.2
5	System	Logs the assignment and mapping events	Audit log entries created for both actions

12.5 Workflow 5 — Client Operator Sends a Permitted Robot Task

Step	Actor	Action	System Response / Validation
1	Robot Operator	Opens the robot's task panel at their site	System shows only robots assigned to the operator's site
2	Robot Operator	Selects a task type (e.g. multi-point delivery) and target point(s)	System restricts task types to those enabled for the robot's vertical (TM-3)
3	Robot Operator	Submits the task	System validates authorization and forwards the task to the Keenon Open Platform
4	System	Receives Keenon callback with task progress	Task status updates in real time; visible to the operator and their organization's admins
5	System	Logs the task submission	Audit log entry created with initiating user, robot, and site

12.6 Workflow 6 — Technician Consumes a Spare Part During Service

Step	Actor	Action	System Response / Validation
1	Technician	Opens the service record for a robot under service	System shows the robot's owning/assigned organization and current stock context
2	Technician	Selects the spare part used (e.g. a wheel motor) and records the quantity	System validates that stock is available within the technician's authorized organization
3	System	Creates a Service Consumption stock transaction and decrements available stock	Reorder-level alert triggers automatically if stock falls below threshold
4	System	Appends the consumption to the robot's Service History	Service record is updated with parts consumed and technician notes
5	System	Logs the transaction	Audit log entry created; visible to the organization's admins per Section 5.2

12.7 Workflow 7 — A Distributor Attempts to Access Another Distributor's Client Data (Denied)

Step	Actor	Action	System Response / Validation
1	Distributor A Admin	Attempts to open a client, site, robot, or inventory record belonging to Distributor	System evaluates the request against Distributor A's hierarchy_path

Step	Actor	Action	System Response / Validation
		B's hierarchy (e.g. via a guessed or shared URL/ID)	
2	System	Determines the target record's organization is not within Distributor A's own organization or authorized descendants	Access is denied with an authorization error (no data returned, including existence confirmation)
3	System	Records the denied attempt	Audit log entry created: actor = Distributor A Admin, action = ACCESS_DENIED, target org = Distributor B's descendant
4	Sakar Admin	May review denied-access attempts in the audit/security report	Repeated attempts can be flagged for investigation

12.8 Workflow 8 — Sakar Admin Reviews Organization-Wide Reports

Step	Actor	Action	System Response / Validation
1	Sakar Admin	Opens the global dashboard	System aggregates robot, task, and inventory data across all organizations (VG/F global scope)
2	Sakar Admin	Filters by organization, date range, robot model, or status	System applies filters without any organization-scope restriction, since Sakar Admin has global access
3	Sakar Admin	Exports a consolidated report	System generates the export and records the export scope in the audit log
4	System	Confirms export completion	Report is delivered via download or scheduled email per user preference

13. Acceptance Criteria

Representative Given/When/Then criteria per functional area. Each maps to the functional requirement IDs in Section 4.

Ref	Given	When	Then
OAM-4	A Sakar Staff account has been granted the “Can create Distributors” flag	That staff member submits a new Distributor creation form	The Distributor is created and appears only to Sakar Admin and that staff member's assigned scope; staff without the flag cannot access the creation form
OAM-5	A Distributor Admin is logged in	They attempt to create a Sub-distributor or Client	The new organization is created as a descendant of their own organization only, and is not visible to any other Distributor
RFM-4	A robot record exists with no Keenon mapping	Sakar Admin links it to a Keenon robot ID already mapped to another Sakar robot	The system rejects the link and displays a duplicate-mapping error
TM-3	A robot is deployed at a hotel site with hotel-delivery APIs enabled, and elevator control disabled	A Robot Operator opens the task panel	Only hotel-delivery task types are available; elevator-control task options are not shown
INV-2	An organization's stock of a spare part is at 5 units	A technician records a Service Consumption of 2 units	Stock is decremented to 3 units and a StockTransaction of type Consumed is recorded
INV-3	A reorder level of 10 is configured for a consumable	Stock falls to 9 units after any transaction	A low-stock alert/notification is generated for the owning organization's admins
SEC-3 / Workflow 7	A Distributor A user is authenticated	They request a resource belonging to Distributor B's hierarchy by direct ID	The request is denied with an authorization error and an ACCESS_DENIED audit entry is created
KIN-5	A Keenon callback is received with an invalid signature	The system processes the inbound webhook	The callback is rejected, not applied to any Task/Robot record, and logged as an invalid-signature event
RPT-2	A Sub-distributor Admin exports an inventory report	The export is generated	Only inventory belonging to the Sub-distributor and its own Clients is included; the export action is recorded in the audit log

14. Delivery Phases / MVP Roadmap

14.1 MVP Scope (Phase 1)

The MVP delivers the minimum feature set required to replace spreadsheet-based operations with a governed, auditable system.

- Organization hierarchy: Sakar Robotics, Distributor, Sub-distributor, Client, Direct Client, with full CRUD scoped per Section 3 and Section 5.
- User & role management with RBAC + OBAC enforcement (Sections 4.1, 4.2, 10).
- Robot registration, assignment to client/site, lifecycle status, and Keenon robot ID linking (Section 4.3).
- Site/customer management (Section 4.4).
- Core Keenon integration: credential management, store/robot/status sync, callback handling with signature validation (Section 7.1, 7.2, 7.5).
- Basic task dispatch: remote call, multi-point delivery, return-to-origin (Section 7.3, excluding vertical-specific APIs).
- Inventory management for robots, consumables, and spare parts with core stock transactions: received, allocation, transfer, installation, consumption, adjustment (Section 8).
- Role-based dashboards and the core report set: Fleet Status, Task Completion, Inventory Stock Level (Section 9).
- Full audit logging of create/update/delete/export/login/denied-access events (Section 10.3).

14.2 Phase 2 Scope

- Vertical-specific Keenon APIs: hotel, hospital, cleaning-zone, and elevator integration (Section 7.3).
- Service History and Stock Audit / cycle-count workflows with variance reporting (Section 8.1, 8.4).
- Reorder-level automated alerts and notification center (Section 4.6, 8.5).
- Advanced reporting: Service History Report, Organization Hierarchy Report, Keenon Sync Health Report, and scheduled report delivery (Section 9.2, 9.3).
- Bulk import tooling (CSV/Excel) for robots and inventory (RFM-7).
- Multi-factor authentication and configurable session/lockout policies (Section 10.4).
- Localization framework for future multi-language rollout (Section 11).

14.3 Phase Summary

Phase	Primary Goal	Key Deliverables
MVP (Phase 1)	Replace manual processes with a governed system of record and safe Keenon connectivity	Org hierarchy, RBAC/OBAC, robot & site management, core Keenon sync, core inventory, core reports, audit log
Phase 2	Deepen operational depth and automation	Vertical Keenon APIs, service history & stock audit, alerts, advanced reporting, bulk import, MFA
Phase 3 (Future, indicative)	Scale and extend	Localization rollout, mobile app for technicians/operators, predictive stock recommendations, partner self-onboarding

15. Open Questions and Assumptions

15.1 Assumptions

- Sakar Robotics is the sole reseller relationship with Keenon; all Keenon API access is brokered through Sakar's own Keenon Open Platform credentials, not per-partner credentials.
- A Client organization has exactly one parent in the hierarchy at any time (either a Distributor, a Sub-distributor, or Sakar Robotics directly for Direct Clients) — no dual-parenting.
- Sub-distributors are at most one level deep beneath a Distributor (no Sub-sub-distributor tier), per the hierarchy defined in Section 3.
- Robots are uniquely owned by one organization at a time; resale/transfer of ownership between organizations is an explicit, audited action rather than a continuous shared-ownership model.
- Keenon Open Platform provides the callback and polling capabilities described in Section 7; exact API contract details will be confirmed against current Keenon API documentation during technical design.
- Initial launch is English-language, web-based (desktop-first), with technician/field workflows supported via responsive mobile web rather than a native app.

15.2 Open Questions

- What audit-log retention period is required for compliance purposes (e.g. 1 year, 3 years, indefinite)?
- Should Distributors be permitted to resell/transfer robot ownership to Sub-distributors within the platform, or is ownership always retained by Sakar Robotics?
- Are there regulatory or contractual constraints (e.g. data residency) affecting where the Platform and Keenon integration data may be hosted?
- What is the expected SLA for parts fulfillment/reordering across the distributor network, and should the Platform automate purchase-order generation in a later phase?
- Should partner organizations be able to have more than one admin user, and if so, how are conflicting simultaneous edits to the same record resolved?
- Is single sign-on (SSO) required for any partner or Sakar-internal users at launch, or is it a Phase 2+ consideration?
- What specific RPO/RTO targets are required for disaster recovery, and who owns the infrastructure hosting decision?

Appendix A: Glossary

Term	Definition
OBAC	Organization-Based Access Control — restricting data access based on an organization's position in the hierarchy
RBAC	Role-Based Access Control — restricting actions based on a user's assigned role
Keenon Open Platform	Keenon Robotics' vendor API platform used to manage and control physical Keenon robots
Hierarchy Path	A materialized representation of an organization's ancestry used to enforce scoped data access efficiently
Direct Client	A client organization that reports directly to Sakar Robotics with no distributor/sub-distributor intermediary
Consumable	A low-cost, high-turnover service item (e.g. filter, brush, cleaning pad)
Spare Part	A higher-value repair component (e.g. motor, sensor, controller, PCB)
Reorder Level	The stock threshold below which a low-stock alert is automatically triggered